A Lightweight AI-Driven Framework For Rapid Intrusion Detection in IoT

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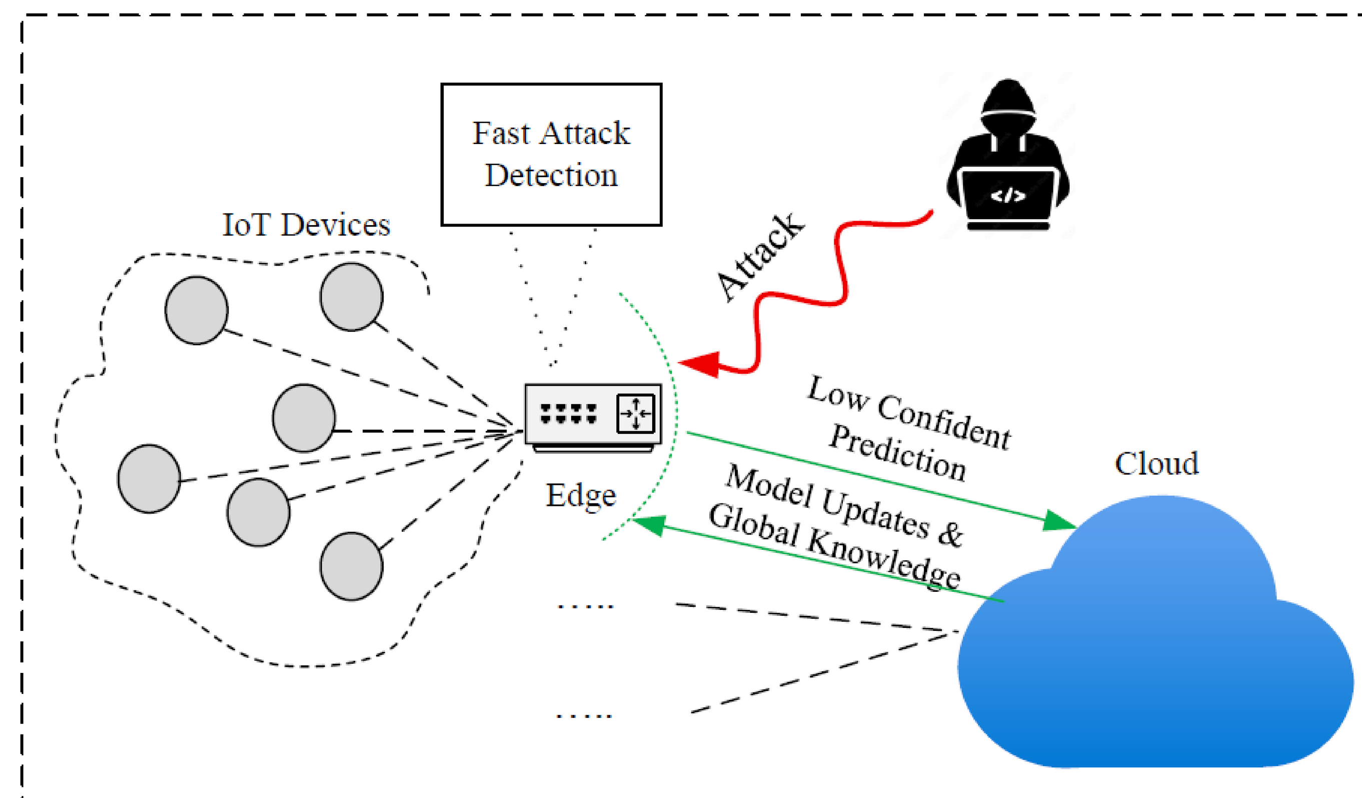
Introduction

- **Background:** IoT systems are ubiquitous but vulnerable. Traditional security fails against sophisticated attacks. Recent solutions use ML or DNN, but face a critical trade-off.
- **The Dilemma**: DNN-based IDS is accurate but too costly for IoT devices. Lightweight ML is fast but misses sophisticated attacks
- **Problems**: Current solutions sacrifice either accuracy or speed. IoT devices lack sufficient CPU, memory, and battery capacity to run complex models. Static IDS designs fail to adapt against evolving attacks and concept drift. High-volume attacks like DDoS overwhelm systems before detection mechanisms can respond.
- **The Gap:** No lightweight IDS that balances accuracy, speed, and adaptability

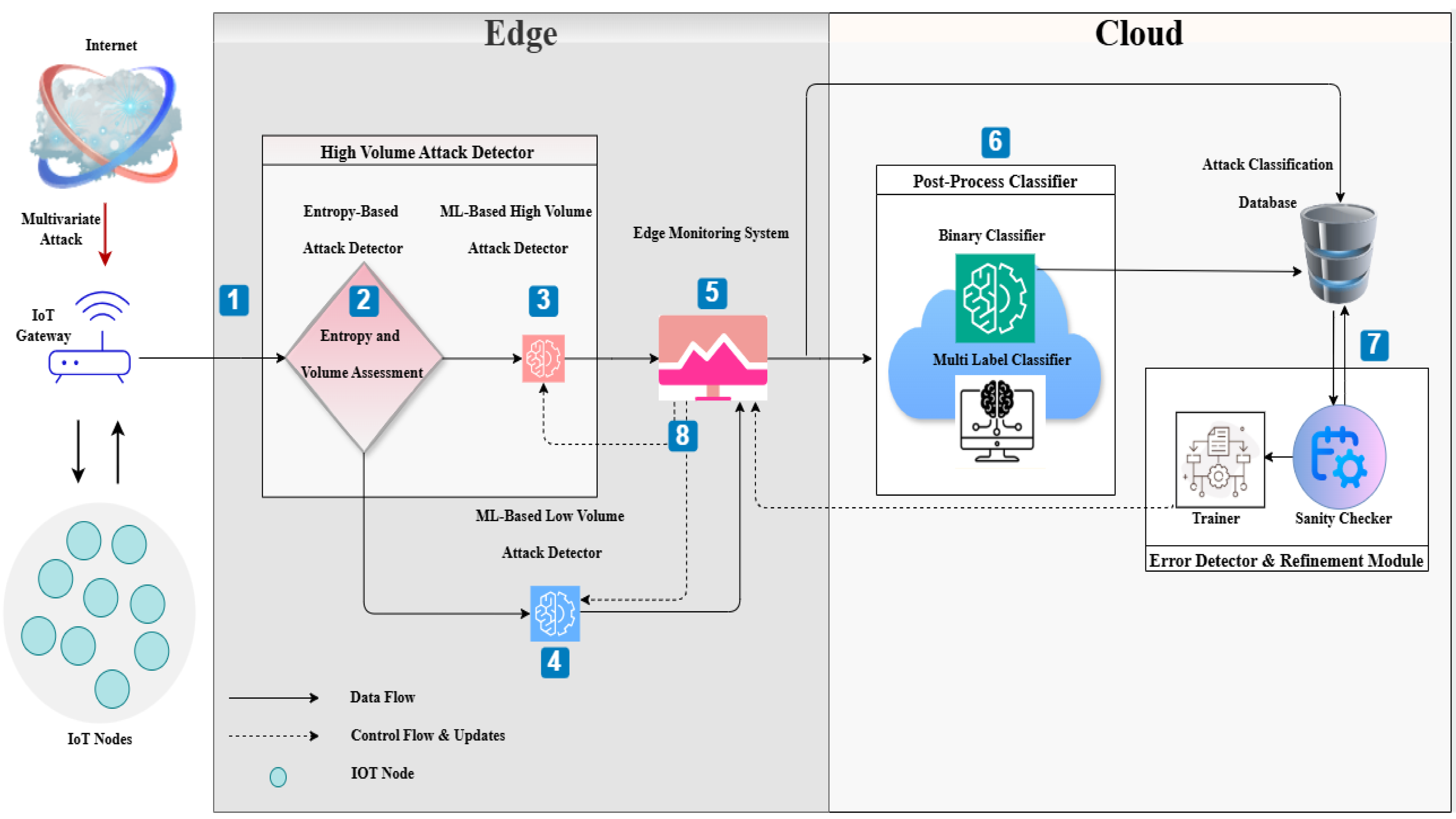
Motivation

Key Challenge: How can we design an IDS that achieves fast detection, high accuracy, and remains lightweight for IoT devices?



Methodology & Framework

- **High-Volume Attack Detector (Two Stage):** Employs an entropy-based detector to identify DDoS/DoS attacks, followed by a lightweight ML-based classifier for secondary validation.
- **Low-Volume Attack Detector:** Employs a lightweight binary classifier (e.g., Decision Tree) to detect non-high-volume attack.
- **Edge Monitoring System:** Performs local retraining and confidence-based cloud offloading.
- **Post-Process Classifier:** High-accuracy reclassification of low-confidence edge predictions.
- **Error Detection & Refinement**: Monitors performance, detects drift, and automatically retrains models.

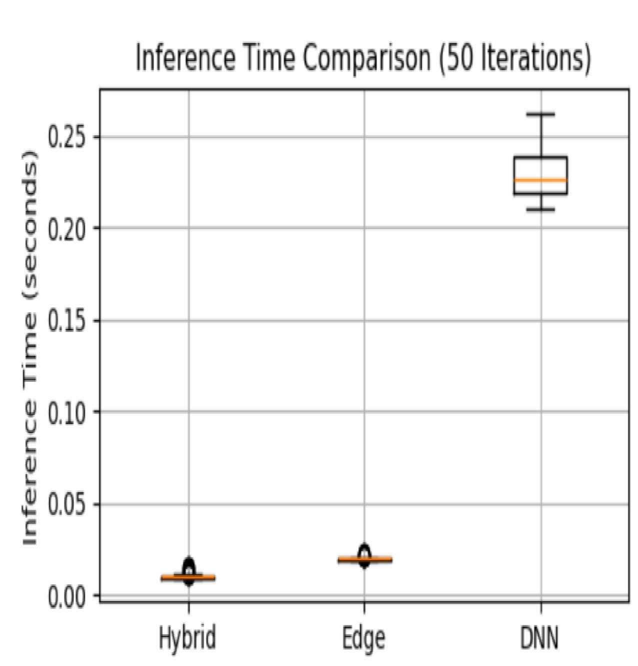


1. The IDS is triggered when attack traffic is detected.
2. The entropy-based module monitors the entropy of destination IP addresses along with the PPS. If the PPS exceeds ε_1 or the entropy drops below ε_2, the traffic is forwarded to the high-volume ML module; otherwise, it is handled by the low-volume attack detector.
3. The ML-based High-Volume Attack Detector re-evaluate the traffic to verify if it was attack, then, it forwards its inference as well as confidence score results to the EMS for further analysis.
4. The ML-Based Low Volume Attack Detector analyzes the traffic to determine if it is a low-volume attack, and then forwards the results to EMS.
5. EMS receives inference and confidence score from prior models. Any inference with confidence score below 50% is passed to post-process classifier for more detailed analysis and also saved in the data base, while high confidence scores are considered as final decision.
6. Post-process classifier re-evaluates the traffic, determine its specific type, and final classification is transferred and saved in database.
7. Sanity Checker periodically fetches the stored results of edge and cloud classifiers, detect misclassifications and passes them to Trainer based on a reactive or proactive strategy. Trainer retrains replicated versions of edge modules using misclassified samples and sends the updated weights and features back to EMS.
8. EMS transmits the updated weights, features and retraining samples to the edge modules.

Performance Evaluation

TABLE I
EVALUATION METRICS FOR THE PROPOSED IDS AT VARIOUS CONFIDENCE THRESHOLDS AND BASELINES

Framework / Confidence	Class 1			W. Avg.		
	P	R	F1	P	R	F1
Cloud Baseline	1.00	0.99	0.99	0.99	0.99	0.99
Edge Baseline	0.87	1.00	0.93	0.89	0.88	0.85
Proposed IDS (0.85)	0.92	1.00	0.96	0.94	0.93	0.93
Proposed IDS (0.95)	0.99	1.00	0.99	0.99	0.99	0.99



Conclusions

The proposed framework balances accuracy, speed, and resource efficiency for IoT security. It achieves perfect attack detection (recall = 1.0) with minimal false positives (precision = 0.92), while outperforming cloud-based approaches and existing baselines in response time. Its lightweight design supports deployment on edge devices, adapts to evolving threats, and remains resilient during network disruptions.